Maya protocol (CACAO) Liquid Thesis

Analysis by Shabbir Khan

Executive Summary —

- Maya Protocol is a cross-chain liquidity infrastructure protocol, forked from THORChain, designed to deliver secure and efficient swaps across native assets without using bridges or wrapped tokens. The ecosystem includes MAYAChain (cross-chain AMM), AZTECChain (smart contract platform under development), and a multi-token system featuring $CACAO, $MAYA, and $AZTEC. With a fully circulating token supply, strong architectural decisions, and THORChain interoperability, Maya positions itself as a capital-efficient alternative with a high-upside beta play to $RUNE.

Token Economics —

- Fully diluted mcap: $58M | TVL: $44M
- No emissions: Yield is entirely from trading fees
- $MAYA yields approx. $2/token/year at current volumes
- ILP starts at 50 days, maxes at 400 days
- Their biggest peer $FLIP just started trading w FDV of 261M.
- $AZTEC:(peer) is a revenue-sharing token limited to the Aztec Chain.
- CACAO looks significantly undervalued.

$MAYA: A rev-share token where holders receive 10% of swap and tx fees from MAYAChain. 1M supply, 15.6% locked forever by code for the team. Earning at around $2 per MAYA per year with current volume

Yield Dynamics & Liquidity Mechanics —

- Yield = Trading fees (100%)
    - 80% to LPs
    - 10% to $MAYA holders
    - 10% to ILP treasury
- CACAO is paired 50:50 in liquidity pools (e.g., BTC/CACAO).
- Strong price correlation with $RUNE due to shared LP pool.

Security & Audits —

- Audited by Halborn Security..
- Utilizes native asset vaults, no custodial bridge risk.
- Economic security enforced via over-bonding (node capital exceeds TVL).

Since it's a THORChain fork, many of its functionalities follow the similar route but since TC has limitations due to its architecture, that gives space for Maya to shine.

- Ledger and Trust wallet prefer THORChain to avoid risks associated with centralized entities controlling customer funds during cross-chain swaps.
- Other protocols take custody of assets during the swap process, releasing either the same or a wrapped version on the destination chain.
- Risks involved include trusting the entity facilitating the swap to return assets and the potential depegging of wrapped assets.
- Depegging can occur if the protocol lacks sufficient original assets to back derivatives or if the market perceives a shortfall.
- THORChain exclusively uses native assets, eliminating the risk of depegging in received assets and facilitating cross-chain swaps in a single transaction without holding user assets.
- Transactions are enabled through liquidity pools, where each integrated chain has a pool with its native asset and $RUNE.
- Pools include $ETH/$RUNE, $BTC/$RUNE, and others, alongside stablecoin pools like $USDC and $USDT.
- The protocol design mandates that nodes bond more $RUNE than the total value locked (TVL) in non-$RUNE assets.
- The requirement ensures that over twice the dollar value of $RUNE is locked compared to the dollar value of non-$RUNE TVL.
- Increased use of THORChain will raise Annual Percentage Rates (APRs), incentivizing liquidity provision, reducing slippage, and fostering more THORChain usage, creating a reinforcing cycle.

Architecture & Components

1. MAYAChain

- Built using Cosmos SDK and Tendermint; secured via GG20 TSS.
- Cross-chain AMM, native asset support, continuous liquidity pools.
- Validator rotation (churning), proof-of-bond system, and Swap Queue for front-running protection.

2. AZTECChain (in development)

- Smart contract layer forked from Cosmos (Gaia).
- Will integrate lending, stablecoins, and NFT tooling.

3. MRC-20 / M-NFT

- MRC-20: Ethereum ERC-20 and Bitcoin Ordinal hybrid.
- M-NFT: Ordinal-inspired NFT standard for collectibles and gaming.

- Used on MayaScan, backed by GLD (Maya Gold) — internal utility token for games and collectibles.
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GLD

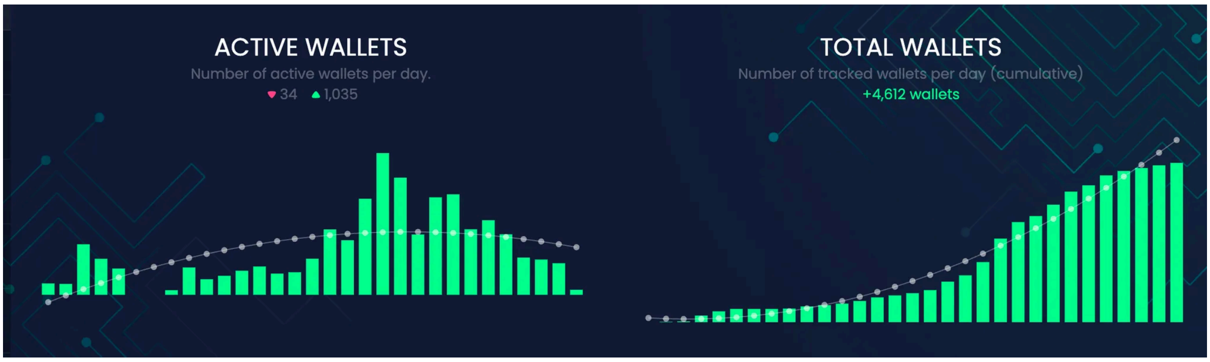
Maya Gold is the main currency of MayaScan; it's used to power blockchain games, buy collectibles and reward users via airdrops and other prizes.

The second GLD airdrop will happen on the 21 Jan ‘24.

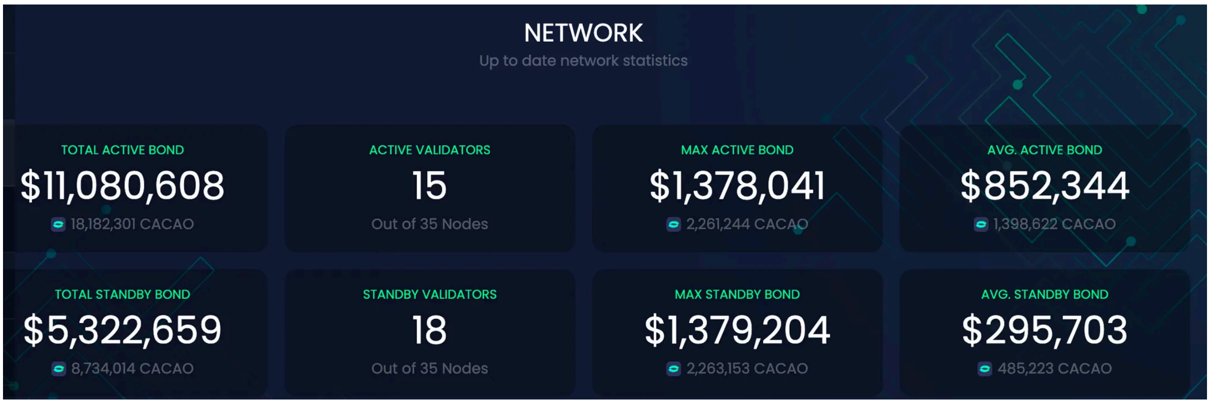
4. M-NFT

- Inspired by Ethereum's ERC-721 and Bitcoin's Ordinals.
- Integrates token deployment, minting, transfer, selling, and buying via memos and ordinal theory.
- Fully indexed and accessible through MayaScan.
- Utilizes memo-resonance tech, similar to ordinal theory.
- Ensures secure, efficient NFT operations, maintaining transaction integrity and digital asset authenticity.
- Reimagines NFT landscape, enhancing engagement for creators, collectors, and enthusiasts with digital art and collectibles.

DAU/MAU growth numbers, revenue numbers



for the past 30 days.



view stats at [Maya Protocol Block Explorer (mayascan.org)]

18% of the supply is with just 35 validators, compare that to TON which has 10% amongst 320 validators. or Thorchain, where 25% of the supply is with 105 validators.

The core economic loop of Maya Protocol revolves around its unique ecosystem, which includes the MAYAChain, the AZTECChain (in development), and its token system.

1. Token System

- $CACAO: Used for transaction fees on both Maya and Aztec Chains.
- $MAYA: A revenue-sharing token where holders receive a portion of swap and transaction fees from MAYAChain.
- $AZTEC: Planned to be a revenue-sharing token for the Aztec Chain, distributing a portion of transaction fees to its holders. A Fork of cosmos, brings smart contracts to life with $CACAO as Gas/Fee Token. Powers the Maya ecosystem -Driving development, Stablecoins, Lending, NFTs, and more.

2. Fee Generation

- Fees are generated from trades executed on the network.
- These fees are primarily derived from transfer fees and outbound fees.
- The fees contribute to Maya's System Income (SI), which is crucial for the protocol's sustainability.
- 100% OF the yield COMES FROM TRADING FEES. People pay to swap assets, and people providing liquidity get 80% of that. $MAYA holders get 10%, impermanent loss fund (in case LPers get negative volatility) get 10%. Meanwhile, THORChain also issues new tokens as rewards.
- How do yields become high? — fee revenue and coin appreciation. Most of it is fees from paying customers. Some of it is from RUNE and CACAO going up in value. If you have 50% BTC 50% RUNE, and RUNE moons, your liquidity position gets great APYs!

3. System Income and Reserve

No inflation, all tokens are circulating since the fair launch. The primary source of Maya's revenue is the fees charged for trades, which contribute to "System Income" (SI). This income is primarily derived from various fees, such as transfer fees and outbound fees.

- System Income: The total revenue generated within the Maya ecosystem from fees.
- System Reserve: A treasury controlled by Maya Nodes, holding the System Income. It's used for various purposes like Impermanent Loss Protection (ILP) for liquidity providers, funding future development, operational expenses, and as an insurance mechanism.

4. Liquidity and Trading

- MAYAChain facilitates cross-chain swaps without the need for wrapping or bridging assets, enhancing security and efficiency.
- Liquidity providers contribute to the ecosystem by adding liquidity to the pools, incentivized by fees and potential ILP compensation.
- To be more specific about liquidity, with both networks, trading fees (or fees + issuance with THORChain) go into the pools when trades happen. The value in the pools goes up. Assets remain balanced 50/50 between the coin (say BTC) and the native token (RUNE/CACAO).

5. Incentive Structure

- Token holders (especially $MAYA and $AZTEC holders) are incentivized through a share in the protocol’s revenue.
- LPs are incentivized through fees and ILP(impermanent loss protection).
- Maya’s capital efficiency is doubled compared to ThorChain. To minimize idle capital, Maya Protocol tweaked this balance in such a way as to allow node operators to provide liquidity with their staked capital.
- Maya’s native liquidity token, CACAO, can be transferred (exported) to secure other sidechains on the protocol. These could be future NFT marketplaces and other EVM-compatible smart contract platforms.

This economic loop aims to create a self-sustaining ecosystem where user activity generates revenue, which in turn is used to incentivize and reward participants, ensuring the protocol's growth and sustainability.

The RUNE/CACAO pool is the largest pool by far right now, so the more RUNE pumps, the higher the CACAO prices go. This also applies to the other assets in the pools as well, but RUNE has more pool depth. CACAO is a high beta play to RUNE in terms of multiples simply because it is a smaller Mcap, and it's now attached to RUNE's price via the deep LP pool.

Impermanent loss protection

- ILP (Impermanent loss protection) attracts liquidity and acts as a stabilizing factor during market volatility, preventing bank runs as liquidity providers (LPers) don't need to withdraw due to fear or FOMO, knowing they'll receive compensation plus APR.
- In the Maya protocol, ILP protection starts after 50 days and reaches full (100%) protection after 400 days.
- LPers who stay for an extended period are likely to be profitable, potentially disqualifying them from ILP. There's a misconception that ILP creates a debt burden for Maya, but the protocol only owes what's in its reserve, and ILP payments cease if reserves are depleted.

Peer analysis —

- Multichain Integration Challenge: Difficult to truly integrate BTC into DEXes; more than just atomic swaps requires innovative design.

- Long-Term Effort: Requires multi-year journey for effective integration; not achievable in short-term, especially not in a month driven by bull market FOMO.
- Maya's team used the existing codebase, yet took about 2 years. Many teams lack the endurance for prolonged development, limiting the emergence of true competitors.
- Existing Players like ***Chainflip and Serai*** started years ago but still haven't launched, making it tough for new entrants to get results right away unless there's a significant technological breakthrough.
- FLIP just started trading. FDV 261M. VC-backed coin, $CACAO is significantly undervalued with better adoption.

Catalysts

- Streaming Swaps (coming soon)
- RUNE integration for DEX aggregation
- Future support for ADA, KAS, ARB, etc.
- Expansion of MRC-20 / M-NFT gaming layer
- More wallet integrations

Risks

- Platform still undergoing development; some swap/withdrawal issues
- Security dependent on node set and GG20 TSS
- If you provide liquidity, you get fees, but could lose money if CACAO price crashes. But if you do Savers(single sided asset exposure using synthetics in Thorchain), you wont lose money. The upcoming Saver will provide a unilateral pledge place for RUNE Holder (33% cap), will be very similar to Thorchain's savers just lower utilization cap. Of course with variable LP yield instead of fixed.
- a friendly ThorChain fork that inherits the chain's security, though HORChain has been hacked previously, It's been really stable and secure, but anything is possible. Bitcoin had fatal exploits, inflation bugs, you name it. Survived long enough to get super-hardened. THORChain will too. It's still young, but doing great. THORChain pools are way more secure than lightning network.
- Listing bad assets like UST or Luna can be catastrophic as they were for RUNE.
- Smart contracts risks, and exploits are always there and one can't ever be full-proof of that, it's something we have seen time and again, even w the most reputed brands.

Shabbir's outlook —

- The balance b/w token incentives and fees generated is important. Token incentives should be good enough to maintain and increase user engagement and liquidity but it will not erode the revenue from fees as no new tokens are minted.
- The protocol needs to balance growth with profitability. Boarding users and liquidity providers through incentives is important for growth
- Subsequent launches and the economic model they adopt ought to change with the launch of AZTEC Chain.

Verdict: IC Passed